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Volume, Risk, Complexity: What Makes Development Finance Projects Succeed or Fail?

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ABSTRACT

This study presents a quantitative analysis of the determinants of success—and failure—of bilateral development finance projects, based on a unique database covering 5,608 project evaluation results. Linking project success ratings to a comprehensive and partly novel set of explanatory factors along the project life cycle, four research hypotheses are investigated, finding that (I) development finance projects with comprehensive financing are (slightly) more successful, in particular budget funds show a significant positive correlation; (II) the project structure does not significantly influence project success; (III) more-complex development finance projects are less successful, in particular in terms of technical complexity and longer implementation duration; (IV) riskier projects are significantly less successful.

JEL classification: C40, F35, O10, O19

Keywords: development finance, OECD DAC evaluation criteria, meta analysis

1. Introduction

Today's world is shaped by global challenges, including climate change, food and water shortages, rising inequality, and an increasing number of conflicts. The consequences of crises are often particularly felt in low- and middle-income countries (LMICs) that have a limited financial capacity to soften their impact. Development finance can be an important remedy, and an increasing volume of official development assistance (ODA) has been committed over the last decade, reaching a total of at least USD 223 bn. in 2023 (OECD 2024).

Knowledge of the overall effectiveness of such commitments has remained inconclusive (Qian 2015). Certainly, there has been a critical advance in the economic analysis of development interventions due to the work of Nobel laureates Banerjee, Duflo, and Kremer, and a corresponding surge in (experimental) impact evaluations (e.g. Olken 2020). These project-level evaluations are key to understanding individual project effectiveness, and the body of evidence produced to date is impressive. In addition, however, there

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is interest in identifying development results systematically across projects and project types, across sectors and countries—in order to understand which implementation factors, program features, and contextual aspects determine (or not) success in delivering intended impacts (e.g. Dollar and Svensson 2000). This is of relevance given the broad portfolios of typically hundreds of development projects that are simultaneously implemented in any particular LMIC.

Several recent studies analyze the role of country- as well as project-characteristics on development success in more detail. Denizer, Kaufmann, and Kraay (2013) use 6,000 World Bank project evaluations and find that success varies more strongly with project-level "micro" variables such as project size, length, and preparation. Feeny and Vuong (2017) conduct a similar systematic analysis using Asian Development Bank (ADB) evaluations, and for example find that larger aid interventions are more successful. Ashton et al. (2023) deepen these investigations by drawing together new World Bank data in combination with a review of the aid effectiveness literature. While including several new covariates—such as intervention design and staff experience—they conclude that additional project-level variables are required to understand project performance.

The study expands this line of research and constructs a novel database of project evaluations, allowing for an in-depth analysis of micro-level factors that correlate with project success. Specifically, the following hypotheses are derived and tested: (I) Development finance projects with comprehensive financing are more successful; (II) better-structured development finance projects are more successful; (III) more-complex development finance projects are less successful; and (IV) riskier development finance projects are less successful. To assess each of the hypotheses, we map a comprehensive set of project characteristics to four explanatory clusters corresponding to the four hypotheses, and correlate the clusters with a standardized measure of project success.

The data are constructed from KfW Development Banks' projects. KfW manages Germany's development finance commitments on behalf of the German Federal Government, the second largest ODA donor worldwide after the USA (OECD, 2023). With a portfolio spanning all economic sectors and most LMICs, KfW's engagement scope is comparable to that of large multilateral donors. Specifically, this study compiles a unique database of KfW project evaluations that comprises 5,608 individual success ratings and that is representative of the institution's entire portfolio across 96 countries. The analytical sample comprises three types of data sources: (a) key variables coded from the hardcopy evaluation reports, including systematic success ratings using the OECD-DAC criteria in the five dimensions relevance, effectiveness, efficiency, impact, and sustainability; (b) KfW operational project data; (c) external data on economic indicators. Conceptually, the structure of the analytical sample thus corresponds to data constructed for quantitative meta analysis (e.g. Card, Kluve, and Weber 2018).

The empirical results provide four main substantive conclusions addressing the four hypotheses. We find weak support for the first hypothesis: There is a statistically significant positive association between project success rating and financial volume—especially for budget funds, and to some extent for larger development projects. Regarding the second hypothesis, the study finds no systematic evidence either to support or reject the hypothesis: Almost none of the covariates describing the project structure show a statistically significant correlation with success ratings. This implies, inter alia, that project implementation works equally well with different partner types, and independent of the number of institutions involved and whether the project is aligned with the focal sectors of KfW Development Bank in the country or not.

The third hypothesis supposes a negative relationship between complex projects and project success. The study finds strong evidence in favor of this hypothesis, as most variables in the complexity cluster are statistically significant: Specifically, longer project durations per se (not delays), longer time spans between mandate—i.e. commitment of funds—and signing of the contract, as well as technically complex project designs are associated with a less successful rating.

The fourth hypothesis conjectures that riskier projects are less successful. Prior to the start of each project, potential risks are forecast by categorizing their severity and mitigation chances. The results show that the higher the rate of eventual occurrence of these ex ante forecast risks, the lower the success rating. This implies that downward risks are correctly identified, but are difficult to mitigate during implementation.

In the debate on development project effectiveness, these findings are particularly relevant because they imply that many of the characteristics that matter most for projects delivering on their intended impact are under the influence of donor agencies and national governments: the overall project volume, risk anticipation and risk management, complexity of the design, and local ownership and integration in the project. Given that research has found project characteristics to correlate similarly with project success across different donors (Bulman, Kolkma, and Kraay 2017; Briggs 2020), the results are informative for the diverse panorama of donor institutions as well as recipient countries.

The paper contributes to the literature in four main ways. First, it constructs a novel database with several key features: by coding more than 30—partially new—covariates describing project characteristics across the project life cycle it provides the most detailed project data to date for one specific donor, to the best of our knowledge. These covariates—e.g. agency type, various measures of complexity and risk—directly address key gaps identified in the literature regarding quality and depth of micro-level information. Second, the granularity of the dependent variable using five dimensions of success ratings—which are systematically measured across countries, sectors, and over time— allows for the identification relevant variation in project success. This provides a more nuanced perspective on the achievements of development finance projects, since previous research has had to focus on the overall project rating only. In addition, the paper can investigate heterogeneous patterns by region and evaluation criteria.

Third, the data allow us to control for, and probe, evaluator- and evaluation-specific effects. Related research partially relies on self-assigned ratings from project leaders heading the project (Denizer et al. 2013; Feeny and Vuong 2017; Ashton et al. 2023). In contrast, all success ratings in the data are assigned by an independent evaluation function, and the independence of evaluator/evaluation characteristics and the assigned success ratings can be empirically tested. Fourth, the study takes an in-depth look at bilateral donor contributions, complementing the systematic analyses of project success determinants for multilateral agencies.

The next section summarizes the relevant literature and derives the hypotheses. The ensuing section provides background information on bilateral development finance and KfW Development Bank project evaluations. The Data and Descriptive Analysis section describes the data in detail and presents results from descriptive and graphical analyses. The Methodology section delineates the estimation strategy, and the Results and the Methodology section present and discuss the meta regression results and robustness, respectively. The last section concludes.

2. Literature and Hypotheses

The literature on the effectiveness of development aid can be placed between two analytical dimensions: At the most aggregate level is the country-level perspective, with studies typically focusing on economic growth as the outcome variable of development aid and integrating other related macro-level variables (see Dreher, Lang, and Reinsberg 2024, for a recent overview). At the most disaggregated level are analyses of specific development interventions, often in the form of (experimental) impact evaluations. These studies focus on a particular development project, and the outcome variables vary with project type.

The focus of the study lies between these two dimensions of the literature, with features of both: The empirical analyses in this line of research are based on samples of project-level evaluations, which—assembled into a data set—allow researchers to address aggregate-level development effectiveness. That is, this literature takes a development-project-focused perspective and analyzes the role of country and, more importantly, project characteristics on development outcomes. The data are typically constructed from a (large) set of ex post evaluations or project ratings, that were assigned by project staff or evaluation experts.

One strand of this literature focuses more on country-level factors and relates them to project performance. Dollar and Levin (2006), for example, emphasize the importance of institutional quality for project success. Similarly, Chauvet, Collier, and Duponchel (2010) find that some projects fare better in conflict-affected countries. Dreher et al. (2013) link project success to politically driven aid allocation using information on World Bank projects. They generally find no significant effect of political influence on project success, but a negative relationship for projects awarded to fragile countries that are part of the UN Security Council. Heinzel, Cormier, and Reinsberg (2023) add to the evidence from international organizations and find that earmarked funding reduces cost-effectiveness and project performance in international organizations.

A second, parallel strand of this literature focuses on the project-level determinants, with the majority relying on data from World Bank projects. Dollar and Svensson (2000) use information on the failure or success of a World Bank reform program as judged by the World Bank's evaluation group and examine project-level factors such as project preparation and supervision time. Similarly, Kilby (2000) relies on World Bank project performance ratings and finds that World Bank supervision, as a project-related variable, improves the performance of World-Bank-financed projects. Denizer et al. (2013) complement World Bank success rate indicators with additional project information, such as characteristics of task team leaders. One of the key results is the importance of within-country variation and project-level information. Kilby (2015) also uses World Bank evaluation data and finds that projects with longer preparation periods are significantly more likely to be successful. Ashton et al. (2023) provide the most recent and detailed analysis with a focus on project manager characteristics and preparation for World Bank projects. They particularly highlight the importance of project-level information per se, but most importantly, project design aspects.

For other international organizations, Mubila, Lufumpa, and Kayizzi-Mugerwa (2000) examine around 150 African Development Bank projects and highlight the importance of project size and a good policy environment. Feeny and Vuong (2017) provide additional information for the Asian Development Bank (ADB) also pointing to the importance of project size. Comparing success information from the ADB and the World Bank, Bulman et al. (2017) confirm the results of Denizer et al. (2013), highlighting in particular that for both organizations, project-level characteristics explain most of the variation in project success, compared to the country context.

The most comprehensive database in this research field has been compiled by Honig (2019), and its most recent version covers 20,686 success ratings of projects financed by 12 donors between 1956 and 2016 (Honig, Lall, and Parks 2023).[1] One key analytical purpose of the data is the comparison across—bilateral and multilateral—donors, and e.g. Honig et al. (2023) investigate the role of access to information (ATI) policies: The study finds, inter alia, that the adoption of ATI policies by agencies is associated with better project outcomes precisely when these policies include independent appeals processes. Given its comprehensiveness and comparability across donors through the holistic measure of project success, the data have been used in several other studies (e.g. Briggs 2020; Caselli et al. 2021).

The paper continues these lines of research, and builds especially on the findings that (a) project-level variables are key to explaining variation in development intervention effectiveness, and that (b) a broader and more detailed set of such project-level variables is required (Bulman et al. 2017; Ashton et al. 2023). Specifically, we formulate four research hypotheses based on the previous literature, which is tested using an expanded, and partially novel, set of explanatory factors.

1 Note that Eckhard et al. (2023) propose a method to extract performance information from the text of the evaluation reports to gain additional insights. In the future, this has the potential to complement the database with information from other donor organizations that do not rely on a standardized rating system.

Hypothesis I (Project financing): Development finance projects with comprehensive financing are more successful

The study hypothesizes that larger projects with comprehensive financing in terms of volume, instruments, and cofinancing are more likely to be successful. The main rationale behind it is that monitoring efforts as well as procurement and general banking regulations are stricter for larger projects. The previous literature has analyzed the importance of the financial volume for project success, for example Ashton et al. (2023) who also summarize the literature on project characteristics in general. Financial volume is significantly associated with project success in a number of studies, but the direction is ambiguous. Feeny and Vuong (2017) find positive results for the ADB, as does Winters (2019) for the World Bank. Honig (2019) reports an overall positive relationship across nine donors for project size. In contrast, other studies remain inconclusive on the role of financing volume (e.g. Caselli et al. 2021), and Denizer et al. (2013) even find a negative correlation. Apart from the overall volume, the alignment between financial commitment and actual disbursement also matters: Feeny and Vuong (2017) find a negative relationship between lower disbursements and project success, a result also supported by Bulman et al. (2017), who argue that unsuccessful projects are already scaled down during implementation. In terms of the financing structure, Winters (2019) finds evidence of a negative relationship between cofinanced projects and project success due to increased transaction costs.

The extensive set of variables allows for the verification of these existing findings, while adding new dimensions of financing structure to the literature. Specifically, we can analyze whether contractual arrangements—loan vs. grants—and national government counterpart contributions influence project success. The latter is a suitable proxy for ownership, an important concept in development financing.

Hypothesis II (Project structure): Better-structured development finance projects are more successful

The study hypothesizes that projects with a higher quality at entry through an—among other features—context-specific design are more successful. Structural design features are, in theory, highly relevant from a policy perspective and could help improve development project effectiveness. There is some evidence that a tailored project design is a determinant for development outcomes on both the individual project (e.g. Khwaja 2009) and aggregate levels (e.g. Wane 2004). Decisions with regard to project structure are taken in the appraisal phase, and are therefore crucial for its eventual success. In addition, it is hard to change project characteristics during implementation that were already decided upon in the appraisal phase. The literature has stressed the importance of high-quality analysis in project appraisal documents for project success (e.g. Corral and McCarthy 2020; Chauvet et al. 2010).

Another part of project appraisal is sector alignment: German development aid has dedicated focal sectors for each of the countries in which projects are implemented. These sectors are to some extent selected for German government policy reasons, but mostly due to analyses on the countries' needs. We construct a variable that identifies projects implemented outside those focal sectors, which can be interpreted as politically motivated projects. It is assumed that these outside-focal-area projects are less successful given the lack of cross-project synergies. Other relevant factors for favorable outcomes include the presence of a country director (Honig 2020), and the type of implementing agency. For example, Shin, Kim, and Sohn (2017) find that projects implemented by non-governmental organizations are more successful. Honig (2020) argues that a local office presence might translate into higher engagement with and knowledge in the country, resulting in more-successful projects. Given the importance of due project preparation, both existing findings are verified and further dimensions of project-structure-related variables are added to the literature. This particularly concerns the role of national counterpart institutions that may have a crucial role for project success. The study analyzes whether previous cooperation with implementing agencies, the number and type of project partners, and their support via technical assistance influences project success.

Hypothesis III (Project complexity): More-complex development finance projects are less successful

The study hypothesizes that more-complex projects—both ex ante and during implementation—tend to be less successful. Existing research has found that a longer project duration is associated with lower project success, hypothesizing that longer projects are more complex and, hence, less successful (Bulman et al. 2017; Briggs 2020). This is also related to delays between approval and the start of implementation, which we refer to as years between mandate and contract (Denizer et al. 2013). Yet the direction of the effect is unclear: While a longer time-length from mandate to contract could be an early indicator for eventually hard-to-manage projects, they could also fare better due to thorough preparation (Deininger, Squire, and Basu 1998; Bulman, Kolkma, and Kraay 2017; Kilby 2015). We add information on three additional aspects to proxy the complexity of a project: a delay indicator, which measures whether project implementation is longer than sector-specific durations; an indicator for technically complex projects, which measures whether a technical expert had to be involved in the project; finally, information on changes in the original theory of change. At KfW Development Bank, a theory of change must be developed for each project at appraisal, outlining the key envisioned outputs, outcomes, and impacts. Changes to this theory of change are made when the project structure changes and could indicate that the project framework was not adequate in the first place or had to be updated to reflect operational adjustments, hinting toward increased complexity and thus potentially lower ratings (Blanc et al. 2016).

Hypothesis IV (Project risks): Riskier development finance projects are less successful

The study hypothesizes that projects with a higher risk of failure will ultimately be less successful. The previous literature has highlighted the importance of staff and management characteristics for project success. Risk categorization and management is an important component of each project manager's work. Project managers typically assign risks to a project at the beginning of the project, and depending on the risk level incorporate potential measures to be taken to reduce the risk during implementation. The previous literature has related project success to supervisory leadership and project oversight (Denizer et al. 2013; Kilby 2000), as well as management experience (Ashton et al. 2023). We proxy for this by including project manager turnover as an explanatory variable in the analysis. Adding to previous research, the study includes information on risk as assessed by the project manager. Therefore, the initial risk that the project manager assigned to the project, whether it occurred, and whether the risk was something that the project manager thought could be mitigated is included.

3. Development Finance and Project Evaluations at KfW Development Bank

3.1. Bilateral Development Finance

KfW Development Bank[2] handles the majority of Germany's official international financial cooperation (FC).

In 2022, for instance, KfW committed EUR 10.9 bn. (USD 11.6 bn.) to developing countries around the world (KfW 2022), making it one of the largest bilateral donors worldwide. The funds mainly stem from the German Federal Ministry for Economic Cooperation and Development (BMZ) and finance projects in Africa, Asia, Latin America, and the Caribbean, and South-Eastern Europe. Sector-wise, these engagements address all areas from agriculture to water supply. The institution's breadth of engagement is thus comparable to that of large multilateral development financiers.

Funds committed by KfW are implemented via projects that comprise dedicated investments. They are designed jointly with and implemented by local—mostly public—agencies such as line ministries, with whom financing agreements are concluded, at times together with international cofinancing institutions. This process entails defining the theory of change (ToC), outlining the results framework of the development finance project, including target indicators for project performance. Once projects are completed, a

2 KfW is a German state-owned investment and development bank.

completion report is conducted—summarizing the project's results from the perspective of KfW project managers. This is followed by an independent ex post evaluation of project success (or failure), taking place approximately three years after project completion.

3.2. Project Evaluations

At KfW, the independent evaluation function FCE (financial cooperation evaluation) is responsible for carrying out project evaluations. A random sample of 50 percent of the projects, stratified by nine sectors, is drawn from all completed projects for each year, and thus is representative of KfW's entire FC portfolio. All ex post evaluations conducted at KfW adhere to the internationally established OECD-DAC criteria. That is, each evaluation systematically assesses the five criteria (a) relevance, (b) effectiveness, (c) efficiency, (d) impact, and (e) sustainability of the given project.[3]

Each criterion is rated on a discrete scale from 6 (best) to 1 (worst), i.e. ranging from "very good" to "highly unsatisfactory." Each evaluation also assigns an overall success rating using the same scale.[4]

From the annual sample of projects, one-third is evaluated by FCE staff, one-third by external consultants, and one-third by seconded colleagues from KfW's operational departments. The governance of every single evaluation, however, lies with FCE. That is, (a) FCE supervises external consultants and seconded colleagues, (b) all reports are peer-reviewed internally within FCE, and (c) the absence of conflicts of interest is ensured: Specifically, any person involved in the evaluation process must not have worked on the project or within the responsible department during its implementation. Each evaluation follows a structured process entailing conceptual design, desk study, on-site visit, and report writing. Summary evaluation reports—with standardized tables of contents—are published on KfW's website.

Each evaluation thus constitutes an expert assessment of project success or failure, following an internationally established methodology. Another benefit from exclusively relying on DAC criteria is that all project evaluations are guided by the same normative framework—independent of regional or thematic focus—addressing concerns that development objectives cannot be compared across sectors (Denizer et al. 2013; Feeny and Vuong 2017). For the purpose of this study, therefore, the use of DAC criteria provides us with a large sample of individual success ratings that were systematically and consistently assigned over the entire sampling period.

In contrast to the ADB and World Bank, KfW's evaluation portfolio does not include self-assessments from operational staff, and it is selected on a strictly random basis. Thus, the data are not prone to selection biases (Kilby and Michaelowa 2019) or to overly favorable ratings assigned by project managers themselves (Bulman et al. 2017; Ashton et al. 2023). Still, even when conducted by a formally independent evaluation function, the autonomy of such bodies can be called into question given that they are based within the institution (Denizer et al. 2013). While the same critique could, in principle, be translated to KfW, several reasons speak against it: First, the director of evaluation is recruited externally from academia, and reports directly and only to the executive board; second, evaluation results are publicly shared; third, the broad set of evaluators (FCE staff, external consultants, seconded operational staff) minimizes potential conflicts of interest; fourth, FCE's methodology is reviewed by an external body, the German institute for Development Evaluation (DEval). Moreover, the study can empirically test the independence of assigned ratings and evaluator characteristics and we discuss this in a dedicated part of the Methodology section.

3 These five criteria were defined by OECD-DAC in the 1990s. A sixth criterion, "coherence," was only added in 2020, and therefore most evaluations in the sample cover five criteria.

4 In general, the overall rating is calculated as the rounded, unweighted average of the five criteria ratings. There is one specific exception, however: If one or more of the three criteria sustainability, effectiveness, or impact are rated as 3 or below, then the overall project cannot be rated higher than 3, independent of the ratings assigned to the other criteria. This particular scenario applying the so-called *knock-out criteria* concerns 37 of the 1,124 project evaluations, or 3.3 percent of the sample.

4. Data and Descriptive Analysis

4.1. Meta Sample Construction and Summary Statistics

The sample is constructed from all $N = 1{,}124$ evaluations of development finance projects that FCE conducted between 2007 and 2021, yielding a sample of $N = 5{,}608$ observations on project success ratings (five ratings per evaluation). This is, to our knowledge, the most extensive and up-to-date database on bilateral financial cooperation evaluations worldwide from a single donor.

It covers projects implemented in a total of 96 LMICs and is representative of KfW's FC portfolio (cf. supplementary online appendix table S1.1). As evaluations are conducted after project completion, different project durations imply that the sample effectively contains development projects that started as early as 1990 and as late as 2019. Each of the 1,124 development finance projects in the sample has a unique ID, allowing us to merge system variables from KfW databases with information coded from evaluation reports, covering rich information on project characteristics (the "micro" variables). In addition, the study combines these data with external statistics on contextual factors in the countries during the time of project implementation (the "macro" variables). Conceptually, the resulting analytical sample combines three types of data sources— (a) key variables coded from the hardcopy evaluation reports, (b) KfW operational project databases, (c) external data on economic indicators—and therefore corresponds to data constructed for quantitative meta analysis (e.g. Card et al. 2018).

4.1.1. Dependent Variable: Standardized Project Success Ratings

The main variable of interest is the individual rating assigned for each DAC criterion on a 6–1 scale for a given project. This allows for the utilization of the granularity of both (a) a broader range of measuring success (instead of a binary indicator success/failure) and (b) the full set of individually assigned DAC ratings instead of only relying on the overall project rating. Supplementary online appendix fig. S2.1 displays the respective distribution of each DAC rating in the data at the project evaluation level ($N = 1{,}124$ each). The majority of overall success ratings (top left) are either "4" or "5," with more than 400 cases each, while only a few evaluations rate projects in the most successful category "6" (44 projects in total). The overall share of ratings with "2" and "3" amounts to 19 percent, or 204 projects. The overall mean rating is 4.21.

The remaining panels for the five individual criteria indicate several patterns: First, the distributions for "effectiveness" and "impact" are rather similar to the overall rating, each with an average rating of 4.34. Second, "relevance" displays the highest share of successful ratings with "5" and "6," and thus the highest overall mean rating (4.85). Third, both "efficiency" and "sustainability" show slightly less successful average ratings, attaining 4.07 and 4.18, respectively.

The core of the analysis uses the pooled sample of these individual ratings. For robustness and comparability with the related literature, two additional outcome variables are constructed: (a) an alternative overall project rating based on an unrounded, unweighted average of all five individual DAC criteria ("arithmetic rating"); and (b) a binary variable indicating whether a given project can be considered successful overall. This is indicated by an overall rating of 4, 5, or 6.

4.1.2. Explanatory Variables: Micro-Level Project Characteristics

At the micro-level, the study constructs more than 30 variables capturing all dimensions of key project characteristics. Specifically, we can distinguish four clusters that are linked to the four hypotheses: (a) *financing* of the project, (b) *structure* of the project, (c) *complexity* of the project, and (d) *risks* for implementation. Supplementary online appendix table S1.2 presents summary statistics for the main variables within each dimension, along with several macro variables and the distribution by sector.[5] The table shows the full sample (column 1) and a stratification by major regions, i.e. Sub-Saharan Africa (SSA, col-

5 For a detailed codebook of all variables used, see supplementary online appendix tables S1.8 and S1.9.

umn 2), Asia/Oceania (3), Europe/Caucasus (4), Latin America and the Caribbean (5), and Middle East and Northern Africa (MENA, column 6). Recall that this sample of project evaluations is representative of KfW's development finance portfolio, indicating that the majority of projects are in SSA ($N = 428$ evaluations), followed by Asia/Oceania ($N = 281$).

The top panel on *project financing* indicates that the "average" development finance project (column 1) has a total volume of EUR 41.7 million, 16 percent of which are contributed by the country counterpart. Cofinancing occurs in 21 percent of projects on average, varying across regions from 11 percent (MENA, column 6) to 30 percent in SSA (column 2). The panel on *project structure* shows that 10 percent of development finance projects are not aligned with the main sectors in a given country at the specific time, and could thus be interpreted as politically motivated projects. The average number of institutions involved in a development finance project is four. An average of 23 percent of projects build on previous cooperation with the same implementing agency, a setting that is relatively rare in the Europe/Caucasus region (5 percent) but rather common in the MENA and SSA regions (31 percent and 29 percent, respectively).

Looking at *project complexity* (third panel), the average project duration amounts to 7 years, ranging by region from 5.7 (Europe/Caucasus) to almost 9 years in the MENA region. These averages relate to the fact that in MENA the execution of projects is delayed in 48 percent of cases, while this is the case for only 12 percent in Europe/Caucasus. The overall average share of delayed execution is 23 percent (column 1). The share of technically complex projects ranges widely from 15 percent in Latin America to 67 percent in Asia/Oceania (average: 48 percent).

At the beginning of the project cycle, project appraisals identify and specify potential risks for project success ex ante. As the fourth panel on *project risks* indicates, the average number of ex ante identified risks is four, with very little variation across regional sample splits. The data also capture to what extent these risks actually occurred during implementation: 55 percent of ex ante identified risks occurred in practice, an average that is somewhat higher in SSA (62 percent), and somewhat lower in Europe/Caucasus (49 percent). Finally, the variable "project manager turnover" relates the total number of project managers in a given project to the project duration in years—implying that at an average of 0.48, the project manager changes every second year.

4.2. Descriptive Analysis

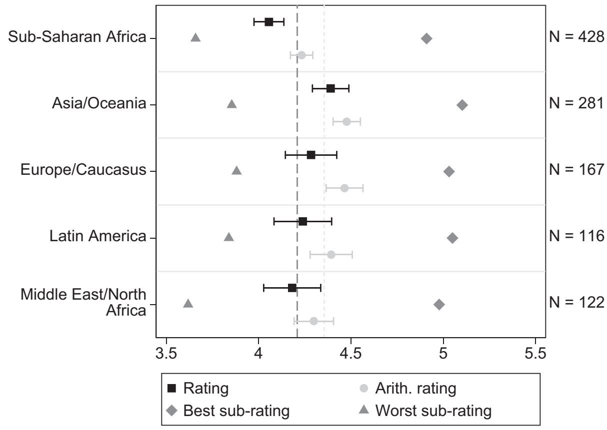
Both the country and the sector where projects are allocated constitute two of the most distinctive project characteristics. Indeed, typically donor institutions are institutionally organized along these dimension. This is also the case for KfW Development Bank and reflects how vital this distinction is for project implementation processes.

4.2.1. Overall Success Rating by Region

Figure 1 shows a forest plot of overall success ratings by region. The blue square represents the average *overall rating*, i.e. the rounded (to a full rating) average of the five criteria ratings that is reported in the evaluation report. The red dot represents the average *arithmetic rating*, i.e. the unrounded, unweighted average of the five criteria ratings. The respective average of the lowest (triangle) and highest (diamond) DAC-criterion ratings are also shown.

The figure indicates several patterns. First, the overall, rounded rating assigned to the project in the evaluation is always lower than the arithmetic mean of the individual criteria ratings. The difference, however, is not very large (4.21 overall vs. 4.36 arithmetic). Second, the respective regional averages are relatively close to the overall averages rather than widely dispersed. However, third, there are some visible regional differences. In SSA, both means are statistically significant below the overall means (4.06 overall, and 4.23 arithmetic). In Asia/Oceania, on the other hand, the mean ratings are statistically significant above the overall means (4.39 and 4.48, respectively). Finally, in SSA and the Middle East/North Africa (MENA) the worst sub-rating (triangle) is consistently lower than in the other regions.

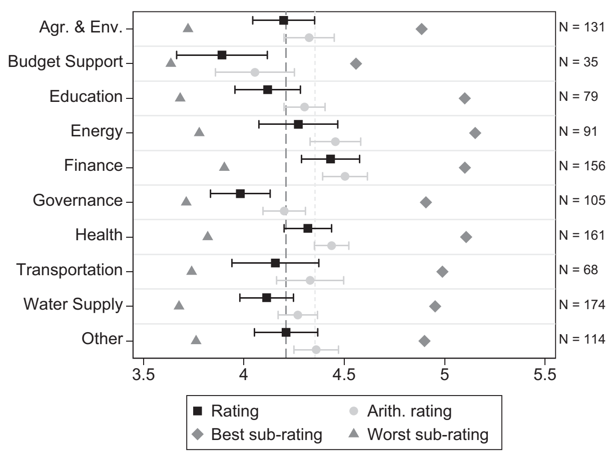
Figure 1. Forest Plot of Ratings by Region.



Source: Internal data and own calculations.

Note: The figure displays mean values of evaluation ratings by region: Black squares denote average overall ratings (i.e. calculated from the rounded unweighted overall rating assigned to a project in the evaluation), light-gray dots denote average arithmetic ratings (i.e. the unrounded arithmetic mean of the five OECD-DAC criteria ratings), diamonds denote means of the highest DAC ratings per project, and triangles denote means of the lowest DAC ratings per project; 95 percent confidence intervals illustrated by whiskers. The black long-dashed and light-gray short-dashed lines mark the sample mean of overall and arithmetic, respectively. Observations are weighted by the inverse number of projects evaluated in the corresponding evaluation report. The *y*-axis on the right-hand side gives the number of observations per category. Ten projects implemented in multiple regions excluded.

Figure 2. Forest Plot of Ratings by Sector.



Source: Internal data and own calculations.

Note: The figure displays mean values of evaluation ratings by sector. See notes for fig. 1. Agr. & env. stands for agriculture & environment.

4.2.2. Overall Success Rating by Sector

Figure 2 displays the corresponding forest plot of average success ratings by main economic sector. The *y*-axis on the right indicates the sectoral distribution of project evaluations and reflects the summary statistic shown in table S1.2: Inter alia, the three largest sectors are finance, health, and water supply, with a share of 15 percent each ($N = 156$, $N = 161$, and $N = 174$ evaluations, respectively).

The figure illustrates notable variation in project success ratings across sectors: Looking at overall average ratings (i.e. blue squares), finance (4.44) and health (4.32) display the most successful ratings, the former statistically significant above the overall average. Projects in the energy sector are also comparatively successful and slightly above average (4.27), with relatively wide confidence bands. On the other hand, budget support (3.89) and governance sector (3.98) lie statistically significantly below the overall average.

5. Methodology

5.1. Empirical Specification

As delineated in the previous section, the meta sample combines rich information on several dimensions of project characteristics with contextual information. Given this structure of the data, the following regression to explain variation in project success is fitted:

$$\text{Rating}_{\text{icrtp}} = \alpha\text{Fin}_{\text{ir}} + \beta\text{Struct}_{\text{ir}} + \gamma\text{Complex}_{\text{ir}} + \eta\text{Risk}_{\text{ir}} + \lambda\text{Eval}_{r} + \theta\text{Macro}_{\text{cp}} + \delta Z'_{\text{irt}} + \epsilon_{\text{ct}}, \tag{1}$$

where $\text{Rating}_{\text{icrtp}}$ denotes the respective DAC-rating dimension of project i, located in country c and evaluated as part of evaluation report r, written in year t. The variables Fin_{ir}, $\text{Struct}_{\text{ir}}$, $\text{Complex}_{\text{ir}}$, Risk_{ir}, and Eval_{r} are vectors of relevant project-specific variables capturing the clusters financing, structure, complexity, risks, and evaluation, respectively. Specific variables within each dimension are discussed further in the results section.

Lastly, the vector Macro_{cp} controls for country-specific characteristics at the time of project implementation p, and Z_{ir} controls for a comprehensive set of additional project-specific variables comprising fixed effects for sector, country, and years of project start and evaluation. Robust standard errors are clustered two ways by country (c) and evaluation year (t). The study estimates equation (1) using weighted least squares (WLS), where the weights are given by the inverse number of projects evaluated in the corresponding evaluation report.[6]

This approach appropriately reflects the research question and data structure, which is similar to meta analysis: A meta analysis draws together a sample of primary (evaluation) studies, all of which address the same or a similar evaluation question using the same or similar methods, in particular using a harmonized or standardizable dependent variable (the "effect size," see e.g. Card et al. (2018) and references therein). The statistical meta analysis then correlates the effect size (in this case, the success rating) with a host of explanatory factors (in this case, the various clusters) to identify systematic patterns.

It has to be noted that the coefficients obtained from estimating the model in equation (1) are prone to endogeneity similar to other related research (e.g. Denizer et al. 2013; Ashton et al. 2023). Development aid responds to macro-economic deterioration and political incentives, which are likely to simultaneously affect the observed outcomes. Despite a rather comprehensive and detailed set of project characteristics, we cannot measure *all* project design features, which in the given context may also respond to unobservable conditions on the ground. Furthermore, finding valid instrumental variables in such settings has proven to be challenging, impeding the identification of causal effects (Bulman et al. 2017; Feeny and Vuong 2017). When discussing the findings, the study therefore points to immediate as well as alternative interpretations that potentially underlie observed estimates. Given that the empirical analysis takes into account an extensive set of fixed effects and control variables, however, we are confident that the results account for unobserved factors to the best of our ability, thereby providing interpretable, relevant, and informative results on the determinants of success and failure of development finance projects, in particular in combination with past studies on the topic (Denizer et al. 2013; Feeny and Vuong 2017; Ashton et al. 2023).

6 WLS avoids oversampling multi-phase projects that are often evaluated in one report with the same grades. Using OLS instead does not significantly change the results as shown in supplementary online appendix table S1.6.

5.2. Independence of Ratings

The analysis benefits from the fact that all success ratings are based on a coherent evaluation methodology. In fact, it is precisely due to the systematic rating framework provided by the DAC criteria—and applied to 1,124 evaluations over 1.5 decades—that it is possible to construct these data (see also Honig et al. (2023)). This coherent, systematic foundation of the data-generation process notwithstanding, there is a possibility that other evaluation-specific characteristics may be significantly related to the assigned outcomes due to potential biases arising in the evaluation process. The study tests for these concerns in turn, and reports the corresponding results in table 1.

First, the sample time lag between the project completion report (i.e. the formal end of the project) and the evaluation report is 3.27 years. This duration might be structurally related to the success rating, since information for projects assessed later might not be as readily available. Also, certain evaluations may only be conducted with delay due to ongoing conflicts in a given country. Such instances might simultaneously affect the outcome. The first row of table 1 reports the corresponding estimation results and finds no statistically significant indication for such a correlation.

A second potential bias concerns the type of evaluator. Whereas all evaluations, ultimately, are conducted under the governance and quality assurance mechanisms of FCE, in practice there are four evaluator categories (cf. background section): FCE staff, seconded colleagues from KfW operational units ("internal"), external evaluation consultants, and internal plus external combined. Ex ante, it is a theoretical possibility that certain types of evaluators systematically assign, on average, too positive or too negative ratings (e.g. it can be plausibly argued that internal evaluators might be tempted to rate too successfully given their expectation that at some time in the future their own projects will also be evaluated). It is one strength of the data that they contain evaluator type information, allowing us to empirically investigate this potential bias. Rows 2–4 in table 1 report the results. Both the reduced (columns 2 and 4) and full specifications (column 5) indicate that the magnitude of the point estimates is small and there is no statistically significant correlation between evaluator type and assigned rating.[7] This is a reassuring finding: The unbiasedness of success ratings is not only plausible given the structural independence of the evaluation function and its evaluators, but is in fact an empirical reality.

A third internal process that could potentially influence ratings is the timing of evaluations during the calendar year. Due to the annual sampling process, the evaluation function has the objective to achieve a certain number of evaluations each fiscal year, and the annual count to achieve that number stops on 31 December. This leads to a clustering of evaluation reports at the end of the fiscal year: 30 percent of the reports in the sample were finalized in December, and 15 percent in November. The remaining 55 percent are relatively equally distributed across the other 10 months. Whereas the pure number of reports per month is no cause for concern, one might conjecture that last-minute reports might potentially be associated with either more positive (in order to finish the report on time) or more negative (the reason why the evaluation took so long) success ratings. Rows 5–15 in table 1 report the corresponding estimation results and indicate that there is no such pattern recognizable in the data, in particular not concerning any end-of-the-fiscal-year pattern. Only April is significantly different from the other months, but it has the lowest number of evaluation reports and simultaneously slightly more positive mean overall ratings (4.45, the remaining months are all between 4.08 and 4.44), such that we interpret this as a deviation at random. Nonetheless, evaluation month as a control variable in all subsequent specifications is included.

A final issue in which institutional evaluation processes might be correlated with success ratings is trends: Over the years, general trends toward better projects—and/or even more ambiguously—better ratings could potentially bias the results. In fact, a slight trend toward better ratings over the sample period is observed; however, mean ratings using five-year evaluation completion brackets from 1990

7 Apart from this regression framework, a one-way ANOVA test also reveals no statistically significant difference between the four groups and the project rating (not shown, available on request).

Table 1. Test for Independence of Success Ratings: Correlation with Evaluation-Specific Characteristics

	(1)	(2)	(3)	(4)	(5)
Time between final review and evaluation	−0.007	–	–	−0.007	−0.022
	(0.012)			(0.012)	(0.019)
Evaluation type (base: FC E):					
—External	–	−0.112	–	−0.135	−0.151
		(0.201)		(0.210)	(0.179)
—Internal + external	–	−0.014	–	−0.018	−0.048
		(0.077)		(0.078)	(0.074)
—Internal (AO)	–	0.038	–	0.029	0.031
		(0.078)		(0.078)	(0.073)
Evaluation month = 1	–	–	0.107	0.120	0.140
			(0.136)	(0.139)	(0.121)
Evaluation month = 2	–	–	0.067	0.067	0.093
			(0.113)	(0.113)	(0.108)
Evaluation month = 3	–	–	−0.039	−0.031	−0.037
			(0.106)	(0.107)	(0.105)
Evaluation month = 4	–	–	0.178**	0.175*	0.229**
			(0.089)	(0.090)	(0.097)
Evaluation month = 5	–	–	0.006	0.009	−0.061
			(0.116)	(0.116)	(0.112)
Evaluation month = 6	–	–	−0.124	−0.125	−0.072
			(0.115)	(0.115)	(0.120)
Evaluation month = 7	–	–	−0.093	−0.088	−0.101
			(0.127)	(0.128)	(0.114)
Evaluation month = 8	–	–	0.033	0.042	−0.044
			(0.093)	(0.093)	(0.098)
Evaluation month = 9	–	–	−0.071	−0.064	−0.072
			(0.117)	(0.117)	(0.104)
Evaluation month = 10	–	–	−0.042	−0.044	−0.085
			(0.099)	(0.099)	(0.098)
Evaluation month = 11	–	–	0.023	0.023	0.010
			(0.079)	(0.079)	(0.073)
Full specification	–	–	–	–	Yes
Sector indicators	Yes	Yes	Yes	Yes	Yes
Country and year indicators	Yes	Yes	Yes	Yes	Yes
Subrating indicators	Yes	Yes	Yes	Yes	Yes
Other control variables	Yes	Yes	Yes	Yes	Yes
Observations	5,608	5,608	5,608	5,608	5,458
Adjusted R^2	0.22	0.22	0.23	0.23	0.30

Source: Author's analysis based on internal data. Supplemented by data from World Development Indicators, Freedom House democracy scores, and the State Fragility Index by the Center for Systemic Peace for full specification.

Note: FC E abbreviates Financial Cooperation Evaluation Unit; ToC abbreviates Theory of Change; WLS abbreviates weighted least squares. Table entries are coefficients from WLS regressions with the pooled rating as dependent variable. Weights are given by the inverse of the number of projects evaluated in the corresponding evaluation report. Other controls include the years of project start and evaluation. The full specification (column 5) includes all variables analyzed in this paper: total volume, aid type, share counterpart contributions, budget funds, disbursement vs. commitment, cofinancing, no sector alignment, accompanying measure, agency type, previous cooperation, number of institutions, country office, project duration, delay indicator, revised ToC, years mandate to contract, technical complexity, number of ex ante risks, share of ex ante risks occurred, overall risks, overall risk control, project manager turnover, GDP percentage growth, population, democracy, and fragility. Standard errors in parentheses are clustered two ways by country and evaluation year. Significance at or below 1 percent (***), 5 percent (**), and 10 percent (*).

onward are not significantly different from one another (not shown in the table for brevity). Nonetheless, all regressions control for year of evaluation.

6. Empirical Results

Previous studies have highlighted the importance of project-level factors to explain the success or failure of development projects for multilateral organizations (see literature section). The results support and strengthen this result for bilateral development aid. As an initial analytical step in relation to this literature, the study calculates the between-country variation in project success and regresses country fixed effects on a binary success outcome variable for each year the projects in the sample were active.[8] From the resulting R^2, the share of variation that can be explained by country factors is derived, i.e. the environment in which the projects are implemented. The result is comparable to that for World Bank projects (Denizer et al. 2013) and indicates that 34 percent (20 percent for the pooled sample) of project variation stems from between-country variation.

In the following empirical analysis, the study thus examines an extensive set of project-level micro variables to contribute toward better understanding determinants of success and failure. Specifically, the following hypotheses are tested: (I) Development finance projects with comprehensive financing are more successful; (II) better-structured development finance projects are more successful; (III) more-complex development finance projects are less successful; and (IV) riskier development finance projects are less successful. To assess each of the hypotheses, the study maps a comprehensive set of project characteristics to four explanatory clusters corresponding to the four hypotheses, and correlates the clusters with the standardized project success ratings.

6.1. Hypothesis I: Are Development Finance Projects with Comprehensive Financing More Successful?

The study utilizes six financial variables such as aid type, share of counterpart contributions, and cofinancing in order to gain a multi-faceted understanding of the role of project finance for project success. The regression results are presented in the top panel of table 2. This table and the subsequent table 3 contain two project characteristic clusters each, and are structured as follows: in the first columns, covariates for the relevant cluster are included in the regression one by one. The second-to-last column reports results from a specification with all variables from that specific cluster. And the last column in each table shows coefficients from the full specification, which includes *all explanatory variables* compiled for this paper.

There is some indication (cluster I) that financially larger projects are systematically correlated with more-successful ratings.[9] The point estimate is positive and statistically significant in the reduced specification (column 1), but becomes insignificant in the full specification (column 8). Financially larger projects may comprise straightforward infrastructure investments or politically prominent showcases receiving more attention, thus making implementation easier.

From a donor perspective, beyond total investment volume, more leverage potentially lies with the budget funds that are committed—i.e. broadly loans vs. grants—as well as the share of counterpart financing contributed by the national government. KfW staff might, for example, be able to exert higher pressure on contractual and regulatory procedures like due diligence when funds are committed as a loan, possibly resulting in better outcomes. Looking at the results in table 2, when compared to grants—which represent 90 percent of development finance projects in the sample—loans do not perform significantly better (cluster I, row 2).

8 See supplementary online appendix S3.3 for a detailed description of the methodology.

9 Recall that the total volume refers to the costs of the entire project, i.e. including commitments by the government itself and/or other donors.

Table 2. Determinants of Success Ratings, Clusters I and II: Project Financing and Structure

Cluster I: Project financing	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Total volume (log)	0.054**	–	–	–	–	–	0.018	0.006
	(0.022)						(0.029)	(0.026)
Aid type (base: loan):								
—Grant	–	−0.013	–	–	–	–	−0.012	−0.022
		(0.092)					(0.099)	(0.087)
Percent counterpart contributions	–	–	0.136	–	–	–	0.108	0.113
			(0.141)				(0.132)	(0.122)
Budget funds (log)	–	–	–	0.095***	–	–	0.082**	0.109***
				(0.031)			(0.038)	(0.037)
Disbursement vs. commitment	–	–	–	–	0.095	–	0.159	0.170
					(0.171)		(0.168)	(0.187)
Cofinancing	–	–	–	–	–	0.022	−0.001	−0.061
						(0.055)	(0.061)	(0.062)
Adjusted R^2	0.23	0.23	0.23	0.23	0.23	0.23	0.23	0.30
Cluster II: Project structure	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
No sector alignment	−0.093	–	–	–	–	–	−0.085	0.003
	(0.078)						(0.078)	(0.070)
Accompanying measure	–	−0.122**	–	–	–	–	−0.122**	−0.096
		(0.062)					(0.061)	(0.059)
Agency type (base: NGO):								
—Mixed	–	–	−0.113	–	–	–	−0.123	−0.177
			(0.119)				(0.118)	(0.126)
—Multilateral	–	–	0.070	–	–	–	0.068	−0.014
			(0.124)				(0.127)	(0.136)
—Private sector	–	–	−0.081	–	–	–	−0.146	−0.140
			(0.139)				(0.137)	(0.136)
—Government	–	–	−0.086	–	–	–	−0.121	−0.153
			(0.098)				(0.097)	(0.101)
Previous cooperation	–	–	–	0.101*	–	–	0.092*	0.093*
				(0.055)			(0.053)	(0.052)
Number of institutions	–	–	–	–	0.011	–	0.013	0.017*
					(0.010)		(0.010)	(0.009)
Country office	–	–	–	–	–	0.065	0.075	0.105
						(0.080)	(0.079)	(0.077)
Adjusted R^2	0.23	0.23	0.23	0.23	0.23	0.23	0.24	0.30
Full specification	–	–	–	–	–	–	–	Yes
Sector indicators	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country and year indicators	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Subrating indicators	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Other control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,608	5,608	5,608	5,608	5,608	5,608	5,608	5,458

Source: Author's analysis based on internal data. Supplemented by data from World Development Indicators, Freedom House democracy scores, and the State Fragility Index by the Center for Systemic Peace for full specification.

Note: ToC abbreviates Theory of Change; EPE abbreviates ex-post evaluation; WLS abbreviates weighted least squares. Table entries are coefficients from WLS regressions with the pooled rating as dependent variable. Percentage budget funds of ODA and percentage projects fund of GDP are re-scaled by 1 million. Weights are given by the inverse of the number of projects evaluated in the corresponding evaluation report. Other controls include the years of project start and evaluation as well as evaluation month. The full specification (column 8) includes all variables analyzed in this paper: time between final review and EPE, evaluation type, project duration, delay indicator, revised ToC, years mandate to contract, technical complexity, number of ex ante risks, share of ex ante risks occurred, overall risks, overall risk control, project manager turnover, GDP percentage growth, population, democracy, and fragility. Standard errors in parentheses are clustered two ways by country and evaluation year. Significance at or below 1 percent (∗∗∗), 5 percent (∗∗), and 10 percent (∗).

Table 3. Determinants of Success Ratings, Clusters III and IV: Project Complexity and Risks

Cluster III: Project complexity	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Project duration (log)	−0.226***	–	–	–	–	−0.199**	−0.075
	(0.080)					(0.081)	(0.084)
Delay indicator	–	−0.055	–	–	–	−0.049	−0.063
		(0.073)				(0.072)	(0.068)
Revised ToC	–	–	−0.084*	–	–	−0.072	−0.088*
			(0.050)			(0.050)	(0.047)
Years mandate to contract	–	–	–	−0.039	–	−0.034	−0.057*
				(0.028)		(0.028)	(0.032)
Technical complexity	–	–	–	–	−0.147***	−0.118**	−0.118**
					(0.057)	(0.057)	(0.055)
Adjusted R^2	0.23	0.23	0.23	0.23	0.23	0.24	0.30
Cluster IV: Project risks	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Number ex ante identified risks	−0.001	–	–	–	–	0.004	0.006
	(0.013)					(0.013)	(0.012)
Percentage ex ante identified risks occurred	–	−0.510***	–	–	–	−0.481***	−0.488***
		(0.071)				(0.071)	(0.068)
Overall risk (base: low)							
—Medium	–	–	−0.273***	–	–	−0.201**	−0.219**
			(0.096)			(0.094)	(0.093)
—(Very) high	–	–	−0.457***	–	–	−0.339***	−0.362***
			(0.103)			(0.099)	(0.101)
—Not assigned	–	–	−0.362***	–	–	−0.254**	−0.254**
			(0.119)			(0.122)	(0.119)
Overall risk control (base: low)							
—Medium	–	–	–	0.033	–	0.054	0.084
				(0.062)		(0.059)	(0.060)
—High	–	–	–	−0.035	–	−0.078	−0.055
				(0.253)		(0.194)	(0.179)
—Not assigned	–	–	–	0.000	–	–	–
				(0.087)		(.)	(.)
Project manager turnover	–	–	–	–	0.406*	0.355	0.410
					(0.218)	(0.226)	(0.255)
Adjusted R^2	0.23	0.26	0.24	0.23	0.23	0.27	0.30
Full specification	–	–	–	–	–	–	Yes
Sector indicators	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country and year indicators	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Subrating indicators	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Other control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,608	5,608	5,608	5,608	5,608	5,608	5,458

Source: Author's analysis based on internal data. Supplemented by data from World Development Indicators, Freedom House democracy scores, and the State Fragility Index by the Center for Systemic Peace for full specification.

Note: ToC abbreviates theory of change; EPE abbreviates ex-post evaluation; WLS abbreviates weighted least squares. Table entries are coefficients from WLS regressions with the pooled rating as dependent variable. Weights are given by the inverse of the number of projects evaluated in the corresponding evaluation report. Other controls include the years of project start and evaluation as well as evaluation month. The full specification (column 7) includes all variables analyzed in this paper: time between final review and EPE, evaluation type, total volume, aid type, share counterpart contributions, budget funds, disbursement vs. commitment, cofinancing, no sector alignment, accompanying measure, agency type, previous cooperation, number of institutions, country office, GDP percentage growth, population, democracy, and fragility. Standard errors in parentheses clustered two ways by country and evaluation year. Significance at or below 1 percent (***), 5 percent (**), and 10 percent (*).

At the same time, the correlation between the share of counterpart contributions and project success is positive, albeit not statistically significant (cluster I, row 3). Intuitively, greater commitment by national governments could be expected to be associated with better ratings, such that this relationship would underline the importance of ownership as a key principle of development cooperation. The overall tendency of a positive association between variables related to the financial volume of a project and project success that can be taken from table 2 is further highlighted by a significant positive correlation between budget funds and project ratings in both the reduced and full specifications (cluster I, row 4).

Donors also decide whether to implement a project along with international partners in a cofinancing arrangement, which is the case for 21 percent of projects in the sample. To increase cooperation is a common pledge among donors, largely due to the assumed positive effects: More streamlined efforts toward developmental impacts and increased efficiency with regard to disbursement conditions have been affirmed in both the Paris Declaration and the Accra Agenda (OECD 2022). The results provide only limited support for this channel, as the coefficient on cofinancing arrangements in table 2 is insignificant and switches its sign depending on the specification.

In sum, the study finds weak empirical support for Hypothesis I with some indication that the overall volume is positively correlated with project success, and a pronounced and statistically significant correlation of budget funds with project success.

6.2. Hypothesis II: Are Better-Structured Development Finance Projects More Successful?

Details of the *project structure* are decided at project appraisal and are at the discretion of KfW. The study includes six dimensions of measuring the quality at entry of the project, the first being whether the project is aligned (or not) with the sector strategy the German development cooperation has in the respective country (10 percent of projects in the sample). If not, the indicator can be interpreted as a proxy for politically driven projects. The study finds no evidence for a systematic difference between projects that are aligned and those that are not (cluster II, row 1).

Development finance projects are often implemented along with technical assistance to support local implementing agencies (27 percent of projects in the sample). A plausible prior belief is that these measures are associated with improved project outcomes. However, the direction is not straightforward, as it could be particularly weak institutions who receive such support in the first place. Such negative selection bias has been argued e.g. to influence the relationship between more diligent project preparation time and unfavorable ratings (Denizer et al. 2013). The estimation results for accompanying measures in the second panel of table 2 are indeed negative and statistically different from zero (cluster II, row 2, columns 2 and 7), a result that lends some support to a negative selection into the need for accompanying measures.

Several more structural design features are worth considering: For instance, donors work with a multitude of local implementing agencies, yet existing research cannot provide detailed insights regarding these institutions' capacities. Increasingly, projects are implemented with non-state actors, responding to the recognition that governmental agencies' capacity is limited (Feeny and de Silva 2012), and potentially allowing for more participatory development partnerships with the civil society. In fact, such projects have been shown to perform better in some instances (Shin et al. 2017). While certain sectors such as microfinance are already dominated by private agencies, most implementing partners in the sample—around 68 percent—are governmental institutions. Distinguishing different agency types, rows 3–6 in cluster II of table 2 find no significant relationship between any of these types and corresponding project success. This is an informative empirical finding for future project design: Agency type is not a significant factor for project success, but previous cooperation as part of other projects with the same institution (row 7) and the total number of institutions involved in a project (row 8) are weakly significant positive predictors. While the former is an intuitive finding—mutual trust and work experience should improve outcomes—the latter is more surprising. Involving more institutions typically implies more complexity, but potentially can address development needs more effectively.

Finally, as a last dimension concerning the project structure, the study explores whether the fact that there is a local KfW representative office in the project-implementing country correlates with the success of a project. The estimation results do not support this assumption.

Overall, the results in this cluster find little support for hypothesis II: While both previous cooperation and the number of institutions are positively correlated (marginally significant) with project success, accompanying measures are negatively correlated. There is no significant association between all other dimensions of project structure and project success.

6.3. Hypothesis III: Are More-Complex Development Finance Projects Less Successful?

The implementation of development finance projects is often complex and challenging. The sample allows for a more detailed investigation in five dimensions of this.

The first dimension of project complexity captures the duration of the project. As row 1 of table 3 shows, for all but for the full specification (columns 1, 6, and 7, respectively), a longer project duration is strongly and significantly correlated with worse success ratings. The eventual duration of a project has both an implementational component, e.g. delays in contracting or executing, and a structural component, as it also depends on the sector or region where it is placed. Row 2 of the table specifically investigates the role of delays, and shows that these are not a significant explanation of lower project ratings.

Concerning another, related factor of complexity, the length of time between the official commitment of governmental funds and their translation into actual projects as part of a contract is theoretically ambiguous. Row 4 in the top panel of table 3 indicates a negative, marginally significant correlation between the length of time from mandate to contract and the success rating in the full specification.

Additionally, we consider whether the ToC (theory of change) outlined at the time of project appraisal was adjusted as part of the evaluation. The study does find a significant negative relationship in the full specification (row 3), implying that these projects fare 0.09 points worse. As a last explanatory factor in this cluster, the study analyzes whether technically complex projects are correlated with better or worse evaluation ratings. This “technical complexity” is a binary indicator variable taking on the value of 1 if the project required the support of a specific technical advisor, e.g. engineers for infrastructure projects. Row 5 of table 3 shows that, indeed, technically complex projects—even when controlling for sector fixed effects—are significantly correlated with less successful project ratings.

The overall finding from the corresponding results presented in table 3, cluster III, confirms hypothesis III: more-complex projects in implementation have a lower likelihood of success.

6.4. Hypothesis IV: Are Riskier Development Finance Projects Less Successful?

A particularly interesting cluster of micro variables in the data is KfW’s internal risk assessment information. Specifically, the data contain information on (a) the number of risks that were identified ex ante (i.e. before project start), (b) the percentage of these that actually materialized during project implementation, (c) the severity of the *overall* risk to project success ex ante (low/medium/high), and (d) the expected level of control of that overall risk (low/medium/high). Furthermore, the study also compiled data on KfW’s internal project manager turnover, which is considered a risk factor.

Row 1 of cluster IV in table 3 presents estimation results for the number of risks identified ex ante. In theory, a larger number implies a more challenging project, yet could also mean that the design is more deliberately thought through to cope with uncertainties during implementation. The results indicate no correlation between the pure number of identified risks and average project success. The key factor that matters for project success, however, is whether and at what rate these pre-identified risks actually materialize: Row 2 consistently shows a strong and statistically significant negative correlation between the share of risks that occurred and the success rating (cluster IV, columns 2, 6, and 7). In fact, the point estimate for the full specification (column 7) implies that projects for which all risks materialize are rated 0.5 points lower. This is a considerable effect size.

Furthermore, rows 3–5 of cluster IV show that high-risk and medium-risk projects are statistically significantly associated with a lower success rating, relative to low-risk projects. Again, the effect is sizable (full specification, column 6): −0.36 rating points on average for high-risk projects, and −0.21 rating points for medium-risk projects, relative to low-risk project. Whether any of these risks was deemed controllable or not ex ante does not affect success ratings (rows 6–8 of the bottom panel). Overall, these findings imply that relevant risks are correctly identified, but are difficult to mitigate—and/or have not been mitigated sufficiently—during project implementation.

In the last row of the bottom panel we estimate the role of project manager turnover. Project managers are in charge of the team at KfW and are the focal point for all interactions with the national government and project implementing unit.[10] Their importance for the success of a project is key, and therefore greater turnover could lead to knowledge loss and thus lower ratings (Ashton et al. 2023). The results indicate an—at first sight—counterintuitive, positive relationship between the number of project managers per year and evaluation outcomes (reduced specification, column 5). Once all other factors in the full specification (column 7) are controlled for, however, this association is no longer statistically significant.

In sum, the empirical results indicate strong support for hypothesis IV, that riskier development finance projects are less successful. And it is of particular relevance for project management that risks can be correctly identified pre-project start, but that they are seemingly difficult to mitigate during project implementation.

Lastly, in addition to the four covariate clusters mapped to the four research hypotheses, we also include country background characteristics, specifically GDP growth, population size, and indices measuring democracy and fragility. Estimation results are presented in table S1.3 and indicate no significant correlation between success ratings and contextual factors.

7. Heterogeneities and Robustness

Disaggregating the results yields further insights and unmasks different patterns within the sample. In addition to the findings for the pooled sample, the study stratifies the analysis by region, and fits separate regressions for the individual DAC criteria.

7.1. Empirical Results by Region

Development institutions regularly identify striking differences in projects' success depending on the region where it was implemented, with Sub-Saharan Africa (SSA) often providing the most challenging environment. Supplementary online appendix table S1.4 disaggregates empirical results by region and shows that, indeed, the correlation between the various determining factors and the project success rating is heterogeneous. In particular, there are only a few variables that play the same significant role throughout all regions, with an exception for variables in the project risk cluster.

Looking at the regions, two variables in the SSA sample (column 2) stand out: Projects financed via grants fare considerably better than loans, potentially explained by the fact that these instruments are used particularly for fundamental public services such as water supply, where ownership could thus be higher. At the same time, projects led by governmental agencies or other institutions are significantly rated worse as compared to NGO-led ones. As the flip side of the ownership argument, it could hint toward (public) institutions' limited capacity when it comes to providing basic infrastructure. This is also supported by the comparatively negative role of technical complexity on project success. Turning to Asia in column 3, projects with higher budget funds are significantly associated with better success ratings, and so are greater disbursement shares of committed funds.

10 Since projects have different durations, the number of project managers per operational year is normalized. A value of 1 therefore indicates that a project was managed by a new manager during each year when it was operational.

Micro variables are more often significantly related with outcomes in Europe (column 4). In particular, covariates in the clusters' project structure and project risks correlate strongly with outcomes, implying that KfW would have a higher leverage to address underlying obstacles ex ante and during implementation. While, for example, long preparation phases before the contract appear to make projects overcomplex, working with an implementing partner with previous KfW experience is associated with considerably better outcomes. Also, involving more institutions in a project contributes to more-successful outcomes in European countries.

Looking at independent variable clusters across regions, several things are notable in table S1.4. First, the point estimate for budget funds is positive almost everywhere, and particularly large in MENA (column 6, financing cluster). Second, in the project structure cluster, the number of institutions involved is positively associated with success in Europe/Caucasus (column 4), but negatively in MENA (column 6). Third, complexity as driven by project duration is a key challenge in Europe/Caucasus and in Latin America (columns 4–5), while technical complexity is a key challenge in SSA (column 2). Fourth, the one coherent pattern across all regions concerns project risks: While the number of pre-identified risks per se is not significant except for MENA, the share of risks that materialize is a significant determinant of project success, or failure, in all regions.

7.2. Individual DAC Criteria

In the next analytical step, main specification for five DAC criteria and the overall rating separately are estimated and results are presented in supplementary online appendix table S1.5. Each criterion addresses a unique dimension of project success and thus provides an additional, detailed perspective on relevant success determinants. While the focus of *relevance* is on the project layout at the time of inception when adjustments are still viable, *efficiency*, *effectiveness*, and *impact* evaluate actual outcomes during implementation. Lastly, *sustainability* concerns outcomes observed at the time of the evaluation, taking into account potential future scenarios of project outcomes. Across the criteria, a first glance reveals that heterogeneities found in the preceding stratifications do not necessarily mirror those at the single criterion level. Nevertheless, only one explanatory variable is consistently significant—the share of eventuated risks—and the other correlates vary considerably.

Given that the success criterion *relevance* (column 2 of supplementary online appendix table S1.5) reflects project design and its ability to address developmental challenges, the *project structure* variables (panel 2) are of particular interest. However, the study finds that none of the micro variables in this cluster, except for the number of institutions, are significantly related to the rating. This also holds for financing variables (panel 1) which are still partially—as in the case of cofinancing—at the scrutiny of KfW. Looking at determinants in the clusters *complexity* and *risks* (panels 3 and 4), the share of risks that materialized displays a significant negative relationship with relevance. Due to the, in theory, structural disconnectedness over time—ex ante relevance of project design vs. actual operational risks materializing—this relationship is somewhat surprising and suggests a level of risk tolerance: KfW correctly anticipates operational risks at the time of appraisal, but from an evaluative point of view these may have already been (partially) rooted in the project design itself. This interpretation is corroborated by the negative and statistically significant relationship between the ex post adjusted indicators of the ToC (panel 3) and the *relevance* outcome.

With regard to *efficiency* (column 3), projects with greater budget funds appear to fare better (panel 1). This potentially stems from large-scale infrastructure investments that undergo more extensive cost-benefit analyses on the part of KfW, rather than projects with regionally spread, small-scale investments. Previous cooperation displays a (marginally) positive influence on project efficiency (panel 2), while technical complexity seems to be detrimental (panel 3). Risk variables as determinants of the efficiency rating show the same pronounced and consistent empirical pattern observed across all OECD DAC criteria (panel 4), and the magnitude of the point estimates is even slightly larger for *efficiency*: both the share of ex ante identified risks that occur and the overall project risk level correlate strongly with lower success ratings.

When assessing the *effectiveness* of interventions (column 4), in addition to the risk pattern found for all outcome variables, the relationship with delays before actual implementation (variable "years mandate to contract" in panel 3)—i.e. time between intergovernmental agreements and the actual project financing contract—is negative. Potentially, this could already constitute a red flag for later implementation challenges, e.g., due to local capacity constraints. This relationship also materializes further down the project logic as displayed in column 5 for the longer-term *impact* criterion. Here, again technical complexity seems to be the hampering factor. At the same time, concerning both *effectiveness* and *impact*, projects with more budget funds are associated with more-successful developmental results.

Lastly, noticeably not a single variable is a positive significant determinant of project *sustainability* (column 6). This is a key result, implying that the already low sustainability ratings—the DAC criterion rated lowest on average in the sample—are difficult to improve from a development institution perspective. In contrast, most risk variables display a negative relationship with the sustainability rating, and so does the actual implementation duration (panel 3). This finding indicates that overly long project periods already hint at difficulties in attaining project objectives, which then in the longer run translate into lower levels of sustainable results.

7.3. Robustness

In the main specification, the outcome variable represents individual DAC ratings that are assigned on an ordinal scale. While estimating the models using WLS allows for straightforward interpretation of the coefficients, the study also estimates equation (1) in alternative specifications to assess the robustness of the coefficients: Supplementary online appendix table S1.6 displays results from OLS and ordered probit models for the pooled ratings (columns 1–3), the overall project rating (columns 4 and 5), the arithmetic rating (columns 6 and 7), and a binary success measure (column 8, see data section) as outcome variables. By and large, the study finds that the coefficients (and significance levels) are comparable across the specifications, and in line with the main results from the preferred specification. Some of the key patterns identified above appear more pronounced for the simple binary success indicator (overall grades 4–6 = "success," grades 1–3 = "failure"), and it also indicates that larger shares of actual disbursement vs. initial commitments determine more-successful projects (column 8).

The selection of the explanatory variables is based on existing theories and past studies. Yet it may still be the case that important, additional variables are ignored, or too much emphasis is put on those variables included. In order to address this issue, the study applies an adaptive Lasso approach. This method automates the variable selection and strips the model to its most predictive variables. More precisely, the variables with the most explanatory power are identified, before equation (1) is re-estimated with the thereby identified reduced set of variables. Results are presented in supplementary online appendix table S1.7. After identifying the variables with the most explanatory power, several control and four main variables in the main specification are dropped. Those are the share of counterpart contributions, indicators for cofinancing and accompanying measures, as well as the delay variable. Re-estimating the model with the reduced number of variables mainly confirms the previous results, as the coefficient size for most variables remains similar. However, two additional variables turn statistically significant: the total volume and the number of project managers in a given project, implying that a greater project manager turnover increases the rating.

8. Conclusion

This paper presents a systematic, quantitative analysis of the determinants of success—and failure—of three decades of German bilateral development finance. The study constructs a unique meta database comprising 5,608 project evaluation ratings, and covering 96 LMICs and all economic sectors. We can empirically test, and establish, the independence of success ratings and evaluator/evaluation characteristics. This is the most comprehensive and up-to-date database on the results of a single donor's bilateral development finance worldwide, and continues and complements a line of systematic research on development aid effectiveness established by, inter alia, Denizer et al. (2013), Ashton et al. (2023), and Honig et al. (2023). The study includes extensive and novel data on project characteristics that allow us to address key gaps identified in this literature and provide new insights into what works in development finance.

Specifically, four main research hypotheses are derived and tested with the following substantive conclusions. First, the study finds weak support for the hypothesis that development finance projects with comprehensive financing are more successful. The empirical results indicate a marginally significant positive association between project success ratings and total financial volume. A stronger and statistically significant pattern is found for budget funds, which are positively correlated with the overall rating as well as the four individual DAC criteria *relevance*, *effectiveness*, *efficiency*, and *impact*.

Second, almost none of the covariates describing the project structure show a statistically significant correlation with success ratings. This implies, among other conclusions, that project implementation works equally well with different agency types, and with different numbers of institutions involved. Another detailed result on project structure is that if a project continues a previous collaboration with an implementing agency, then this has a marginally significant positive correlation with overall success, and a significantly positive correlation with project *efficiency*.

Third, more-complex projects are on average less successful. Several dimensions of project complexity display a statistically significant negative correlation with success ratings: Longer project durations (i.e. implementation durations from contract signing until final review), longer periods between mandate—i.e. commitment of funds—and contract signing, and technically complex project designs are associated with a less successful rating. While project duration and degree of technical complexity are often difficult to adjust (in particular for infrastructure projects), a prolonged time span between mandate and contract could thus serve as an early indicator of a higher risk of project failure.

Fourth, project risk is a key factor associated with project success and failure. In fact, the covariates in the *risk* cluster display the most pronounced, consistent pattern across all specifications and success indicators: the higher the rate of eventual occurrence of the ex ante predicted risks, the lower the success rating. And the higher the overall risk level ex ante, the lower the success rating. This implies that risks have been correctly identified, but are difficult to mitigate and/or have not been sufficiently mitigated during implementation.

In the debate on development project effectiveness, these findings are particularly relevant because they imply that many—but certainly not all—key project characteristics matter for project success, and, moreover, that several among those that matter are under the influence of donor agencies and national governments. These include, specifically, the overall project volume, risk anticipation and risk management, complexity of the design, and local ownership and integration in the project. We would recommend that future project design pay particular attention to these features.

Data Availability Statement

The data underlying this article is constructed from public databases, information publicly available at the KfW Development Bank website, and confidential KfW Development Bank project data. Any data request should be adressed directly to the authors.

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